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Prerequisites

Download all necessary files for this demo from the github repository via this link

<https://github.com/BRERAS/SAS-LS-Analytics-School/archive/refs/heads/main.zip>

Unzip the downloaded file **main.zip** to a folder on your local computer.

Business Context

You are a biostatistician and you work for an important life science company, which is about to launch a new product SASULIN on the market.

On that regard, the company need to conduct a clinical trial to ensure our products safety and approval by the regulating authorities.

In your role as a biostatistician, with no programming skills, your task is to analyze the given data from the clinical trial and answer important questions about the safety of the new product SASULIN.

Business Goal / Questions

Reporting of the Patient Demography Distribution

As a project team we want to understand the distribution of the demography of our patients

Our SAS Statistical programmer just shared with us the new dataset, based on latest data from our EDC. File format is almost CDISC SDTM format (merged datasets)

- How many patients are engaged / enrolled in that clinical trial ?
- What is the number of patients by type of drugs?
- What is the split of patient by age category?
- How is gender impacting the number of patients?
- What are the impacts of race or site number?

Adverse Events this Clinical Trial is potentially causing

In this second dashboard we want to increase our understanding on the number of Adverse Events this Clinical Trial is potentially causing.

We want to create a list of patients facing Adverse Events, gathering Patient ID, Age & Sex. On this list we want to add some important milestones of the Patient during the Trial

- **RFSTDTC** and **RFENDTC** indicate a patient's initial and final exposure to a study drug or treatment, crucial for analyzing patient involvement and events within the study. Understanding the patient's study involvement timeline is crucial for event analysis.
- **RFXSTDTC** and **RFXENDTC** variables specifically represent the date and time of the first and last study exposure to the protocol-specified treatment or therapy, respectively.
- The earliest date of informed consent among any subject (Date/Time of Informed Consent, **RFICDTC**) that enrolled in the study.
- Final date on which data was collected for any subject (Date/Time of End of Participation, **RFPENDTC**) based on the protocol.

For the team this is also critical to understand how those Adverse Events are Across

- Severity (**AESEV**)
- The body system (also know as system organ class) that is affected – **AEBODSYS**
- Event Codification
 - **AEHLGT** coded using the Medical Dictionary for Regulatory Affairs (MedDRA). Sponsors should code each term to the MedDRA-specified hierarchy of terms
 - **AEDECOD** dictionary-derived text description; equivalent to the Preferred Term (PT in MedDRA). The sponsor is expected to provide the dictionary name and version used to map the terms utilizing the define.xml external codelist attributes.
- For that purpose, you will have to create a data hierarchy including all those variables. We will use that hierarchy to count number of Distinct Patients, and it will help the team to look into the data.

The team also wants to understand the investigator's opinion as to the causality of the event to the treatment – **AEREL**

- They ask you to split the number of patients based on the opinion on the causality.
- You should have to graph that you will filter based on the treatment the patient is following. (1 for placebo, 0 for non-placebo). You should use the filtering option “Graph Object”.

- If the patient is treated with treatment (not placebo), the value should be “Green” if the opinion is not related, and orange in other cases (you should use the “Display Rule” option of the object you use).
- If the patient is treated with placebo, if more than 3 events occur, the background of the graph should be colored.

Finally, you will be asked to filter the Dashboard based on the following information:

- Whether the Adverse Event was qualified as serious or not (AESER)
 - The site number (SiteID)
-
- Which patients facing an Adverse Event?
 - How does the event are across?
 - What is the investigator’s opinion as to the causality of the event to the treatment?
 - What Adverse Event’s are qualified as serious or not?

Predict diabetes using several algorithms

At the same time, we have just received from our laboratory experts a specific analysis on SASULIN on Diabetes.

The Diabetes dataset has been shared with our team. As business lead of the Clinical Trial team we want to understand and predict if it has impact on our patients. The variable we are focusing on is: “Outcome”. Be cautious, as the “Outcome” variable can be set as “Measure” variable at first glance. In this example, “Outcome” should be considered as a “Category” variable.

Our first look at the data would be to automatically get from the system what are the main explanatory variables on our main "Outcome" variable; by building an automated-explanation dashboard.

Based on previous results, we want to make sure your mode is optimum or see if there is no way to improve the performance of your Open Source model. Indeed, only two variables were significant. Using the already developed pipeline, you decide to improve it and duplicate it as an intermediate pipeline. You want to check if adding pre-processing node could be beneficial.

- The first test you want to perform is to see if automated “Variable selection” note will be beneficial.
- The second test you want to run requires only the variables that were significant from basic pipeline results.

In everyday work data scientists and/or statisticians may want to compare how different models perform in order to use the best one.

First of all, while discussing with your colleagues that did data exploration on your diabetes datasets, some data anomalies were noticed:

- bloodpressure values equal to 0
- outliers in the pregnancies variable (more than 10)

As an experienced data scientist, you know those outliers can impact your model prediction. You want to remove them from the analysis. You decide to improve the previous model, by duplicating the intermediate pipeline into an Advanced one

As you have been experiencing different steps in your model journey, and on the quality of the model you have developed, you are now at a stage you want to select the optimum models by comparing the different pipelines.

As a conclusion, among all the models of the different pipelines which is the best model?

Survial Analysis

Survival analysis is a branch of statistics for analyzing the expected duration of time until one event occurs, such as death in biological organisms and failure in mechanical systems. This topic is called reliability theory, reliability analysis or reliability engineering in engineering, duration analysis or duration modelling in economics, and event history analysis in sociology. For the following tasks, we will be focusing on terminology that is appropriate in a context of a clinical trials study (e.g. patient's death/survival). This analysis attempts to answer certain questions, such as:

- What is the proportion of a population which will survive past a certain time?
- Of those that survive the main event, at what rate will they die?
- Can multiple causes of death be taken into account?
- How do particular circumstances or characteristics increase or decrease the probability of survival?

Data Dictionary

Four data sets, contains several information and observations about the patients who took part in the clinical trial. The data are available in ADAM and CDISC SDTM format, a data format established in the clinical environment.

Table: **ADTTE**

Description:

26 Variables, 254 Rows

Excerpt

Variable	Description
AVAL	is the elapsed time to the event of interest from the origin. For example, if AVAL is measured in days, AVAL would be ADT – STARTDT or ADT – STARTDT + 1
CNSR	is a required variable for a time-to-event analysis dataset, though it is a conditionally required variable with respect to the ADaM BDS. For example, CNSR = 0 for event and CNSR > 0 for censored records.
TRTP	is a record-level identifier that represents the planned treatment attributed to a record for analysis purposes.

Table: **DM_DM**

Description: Demographic Data

28 Variables, 60 Rows

Excerpt

Variable	Description
USUBJID	Patient ID
ARM / ARMCD	planned treatment the subject taken (ARMCD if you want to use the code instead of the label)
ACTARM/ACTARMCD	actual treatment the subject taken (ACTARMCD if you want to use the code instead of the label)

RACE	Race
SEX	Gender
SITEID	Site Number

Table: **AE_DM**

Description: Adverse Events

54 Variables, 40 Rows

Excerpt

Variable	Description
RFSTDTC / RFENDTC	indicate a patient's initial and final exposure to a study drug or treatment, crucial for analyzing patient involvement and events within the study. Understanding the patient's study involvement timeline is crucial for event analysis.
RFXSTDTC / RFXENDTC	variables specifically represent the date and time of the first and last study exposure to the protocol-specified treatment or therapy, respectively.
RFICDTC	The earliest date of informed consent among any subject (Date/Time of Informed Consent) that enrolled in the study.
RPENDTC	Final date on which data was collected for any subject (Date/Time of End of Participation, RPENDTC) based on the protocol.
PATIENTID	Patient ID
AGE	Age
SEX	Gender
AESEV	Severity
AEBODSYS	The body system (also know as system organ class) that is affected.

AEHLGT	coded using the Medical Dictionary for Regulatory Affairs (MedDRA). Sponsors should code each term to the MedDRA-specified hierarchy of terms
AEDECOD	dictionary-derived text description; equivalent to the Preferred Term (PT in MedDRA). The sponsor is expected to provide the dictionary name and version used to map the terms utilizing the define.xml external codelist attributes.
AEREL	Adverse Event Relativity

Table: **DIABETES**

Description: Linked patient data between treatment and Diabetes

9 Variables, 768 Rows

Variable	Description
OUTCOME	Binary, does the drug have an impact on the patient.
AGE	Age
BloodPressure	Blood Pressure
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes Pedigree
Glucose	Glucose
Insulin	Insulin
Pregnancies	Pregnancy
SkinThickness	Skin thickness

Access the SAS Software - SAS Viya for Learners

Pre-requirements:

1. You created a SAS User Profile with your **university email address**
(<https://www.sas.com/profile/ui/#/create>)
2. You enrolled to SAS Viya for Learners
(https://www.sas.com/en_us/software/viya-for-learners.html)

A detail step-by-step description for the necessary step scan be found in the document
“**registration-guid-skill-builder-for-students.pdf**”

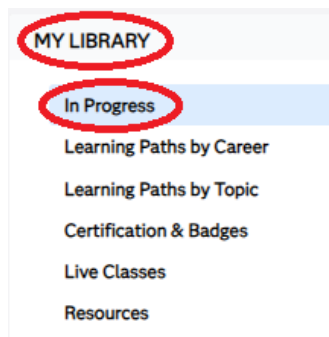
Step 1: Navigate to the following URL

<https://learn.sas.com>

and select **My Training** and use your Username & Password (defined by you, when you created a SAS User Profile)



Step 2: Navigate to **MY LIBRARY** on the left-hand side and select **In Progress**



Step 3: In the middle of the screen, you see all courses you are enrolled. Select **SAS Viya for Learners**

IN PROGRESS



Step 4: Select **Launch Software** and the **SAS Landing page** will open

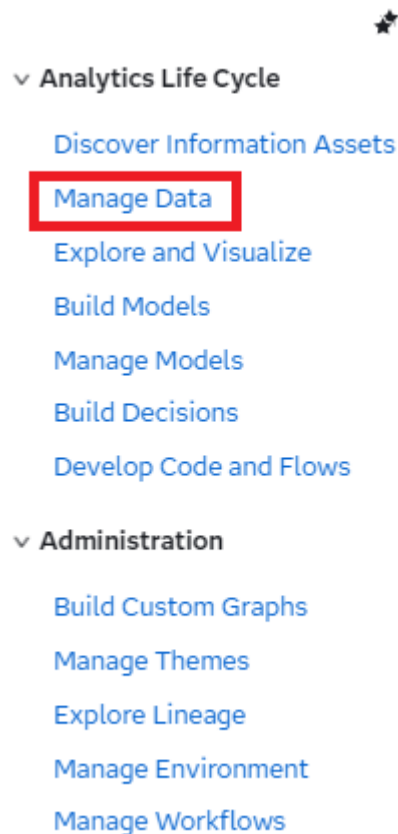
SAS Viya for Learners 4



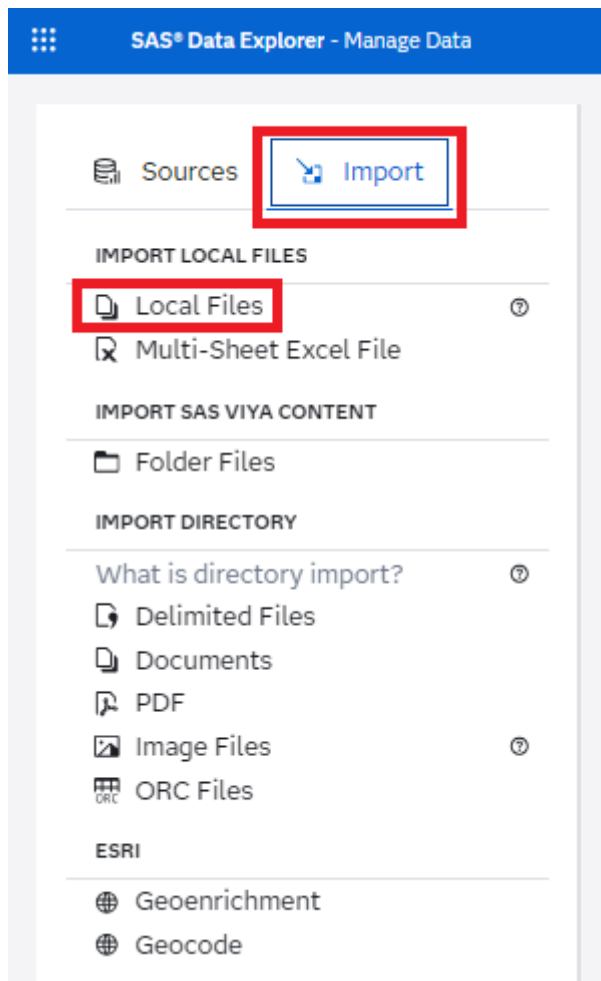
Manage Data

Step 1: Select the '**Applications menu**'  on the left-hand side (left top corner).

Step 2: In the menu panel on the left-hand side choose '**Manage Data**'.



Navigate to the tab “Import” and select “Local Files”



to upload the following files

ADTTE.csv
AE_DM.csv
DIABETES.csv
DM_DM.csv

These are comma seperated text-files.

For each table select the **CASUSER(<your Email>)** as the **Location (library)**, because this is the only library with write access.

DM_DM.csv

> Step 1 - File Import Options

▼ Step 2 - Output File Options

Name: *

Location (library): * [Go to data](#)

Label:

Format:

If table name exists:

☒ Cancel import

☐ Replace file

☐ Save as in-memory data only

Select “Import all”

SAS® Data Explorer - Manage Data

Sources Import (4)

Queue (4)

- DM_DM.csv Local file
- DIABETES.csv Local file
- AE_DM.csv Local file
- ADTTE.csv Local file

DM_DM.csv

Import item **Import all**

Delimiter: Comma

> Step 1 - File Import Options

▼ Step 2 - Output File Options

Name: *

Location (library): * [Go to data](#)

Label:

Format:

If table name exists:

☒ Cancel import

☐ Replace file

☐ Save as in-memory data only

Navigate to the tab “Sources” and select “cas-shared-default” (on the left hand side of the screen). Select **CASUSER(<your Email>)** under cas-shared-default and notice the tables (permanent data or In-memory data) available. Select the table named **DM_DM** (In-memory table). On the top right-hand side you will see information about number of columns, number of rows and the size of the table. Under the tab “Details” you see the variables and there informations e.g. type, label, format and under the tab “Samples” you can see an preview of the data itself.

DM_DM

Columns: 28 Rows: 60 Size: 33.4 KB

Details Sample data

Filter

#	Name	Label	Data Type	Raw Length	Formatted Length	Format
1	STUDYID	--	varchar	6	6	--
2	DOMAIN	--	varchar	2	2	--
3	USUBJID	--	varchar	12	12	--

DM_DM Details Sample data Columns: 28 Rows: 60 Size: 33.4 KB Actions

60 of 60 Rows Sample rows: 100

STUDYID	DOMAIN	USUBJID	SUBJID	RFSTDTCT	RFENDTCT	RFSTDTCT	RFENDTCT	RFCDTCT	RFENDTCT	DTHDTC
XYZ123	DM	XYZ123100101	101	2010-04-02	2010-10-10	2010-04-02	2010-10-10	2010-03-28	2010-10-10	
XYZ123	DM	XYZ123100102	102	2010-02-13	2010-08-10	2010-02-13	2010-08-10	2010-02-07	2010-08-11	
XYZ123	DM	XYZ123100103	103	2010-05-16	2010-11-14	2010-05-16	2010-11-14	2010-05-12	2010-11-15	
XYZ123	DM	XYZ123100104	104	2010-01-02	2010-07-04	2010-01-02	2010-07-04	2009-12-28	2010-07-04	

If the data are not available as In-memory table , then right click on the permanent data  with the sashdat extension **“*.sashdat”** to load the data into memory.

Name	Columns
 DM_DM.sashdat	

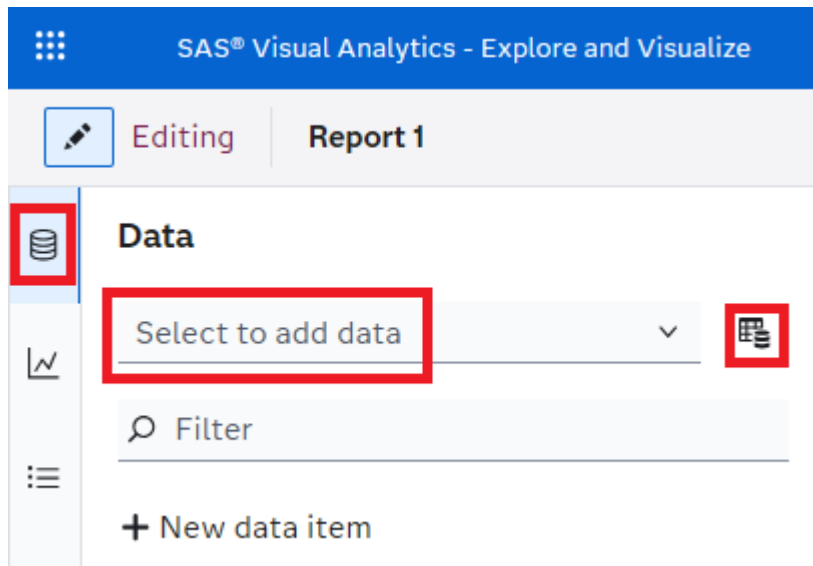
View data
Actions
Load into memory
Add to import
Delete

Explore Data

Step 1: Navigate again to the ‘Applications menu’  on the left-hand side (left top corner).

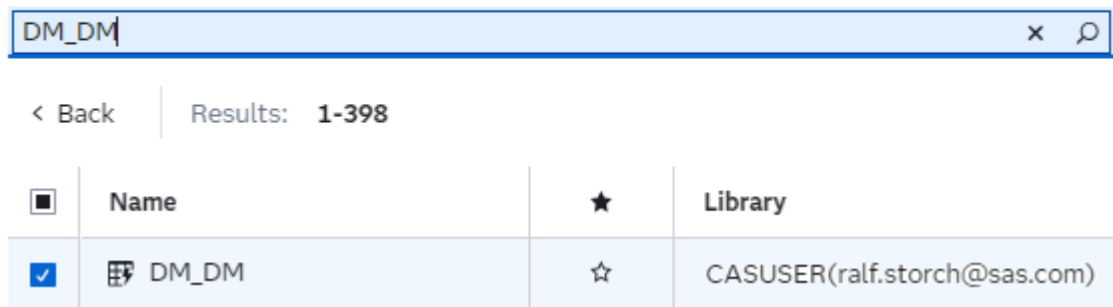
Step 2: Select ‘Explore and Visualize’ and select ‘New Report’  in the top right corner.

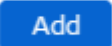
Step 3: Choose the ‘Data’  option on the left-hand side and click inside the field ‘Select to add data’ or click on the ‘Data source menu’ icon to the right of the input field and select ‘Add data’.



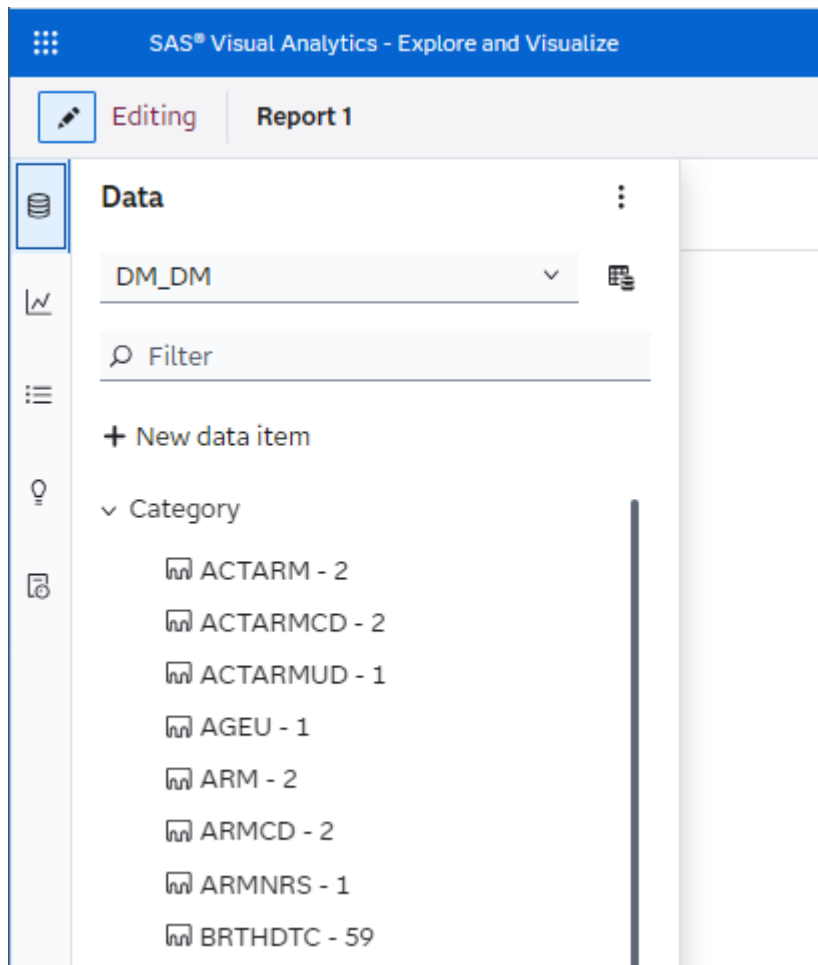
Step 4: Search for “DM_DM” in the search text box and press the Return key or click on the magnifier glass. “DM_DM” data will appear on the screen.

Choose Data



Step 5: Select (click once on the) “DM_DM” data already pre-loaded into memory. Hit the ‘Add’ button (bottom right corner) .

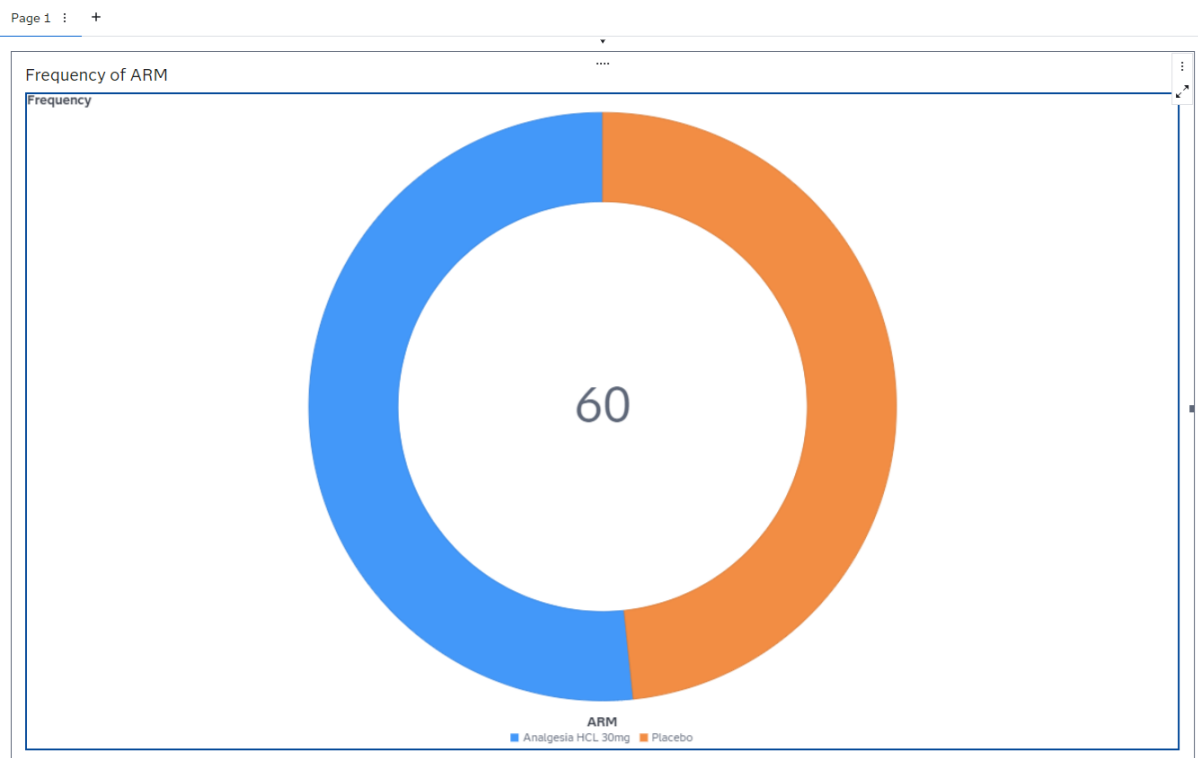
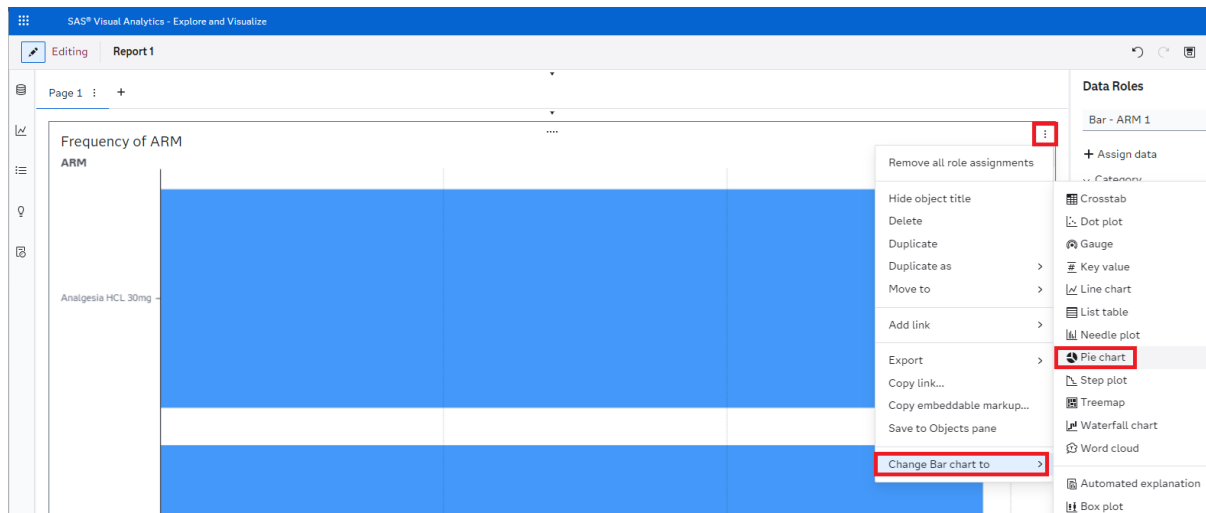
You will now notice that this data is available for data visualization.



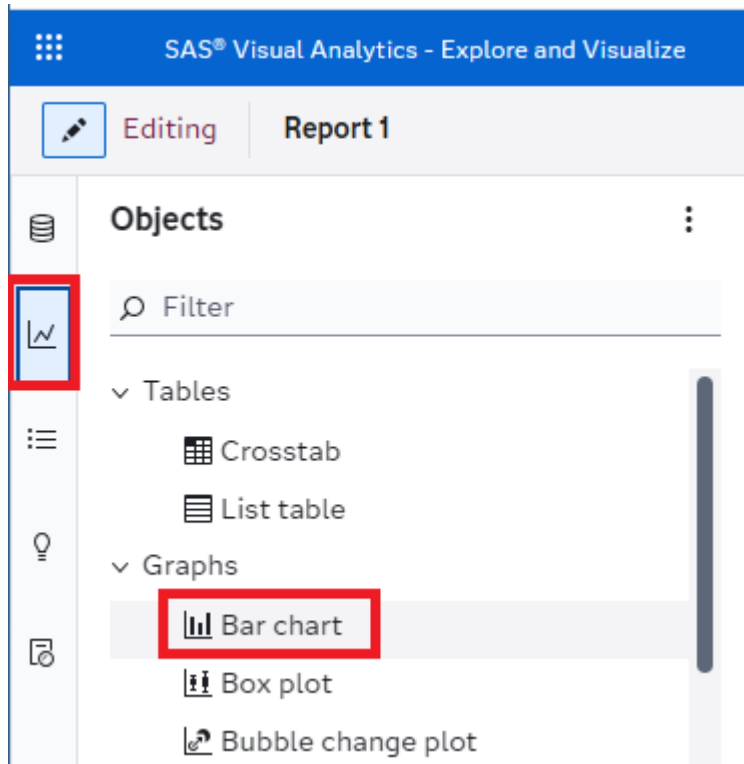
Demographic Dashboard

Step 1: Drag & Drop the Category variable **ARM** on the canvas.

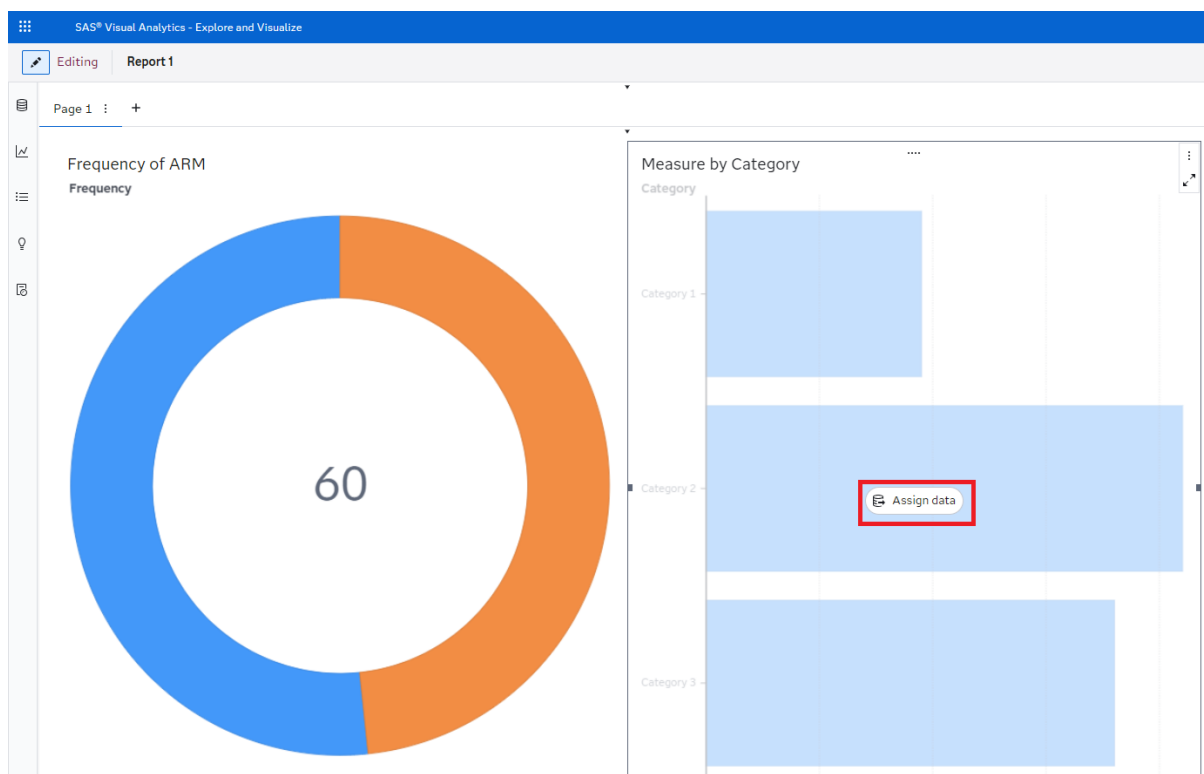
Step 2: Change the **Bar chart** to a **Pie chart** by selecting the object menu of the Bar chart (3 vertical dots in the top right corner of the Bar chart) and select Change Bar chart to > Pie chart



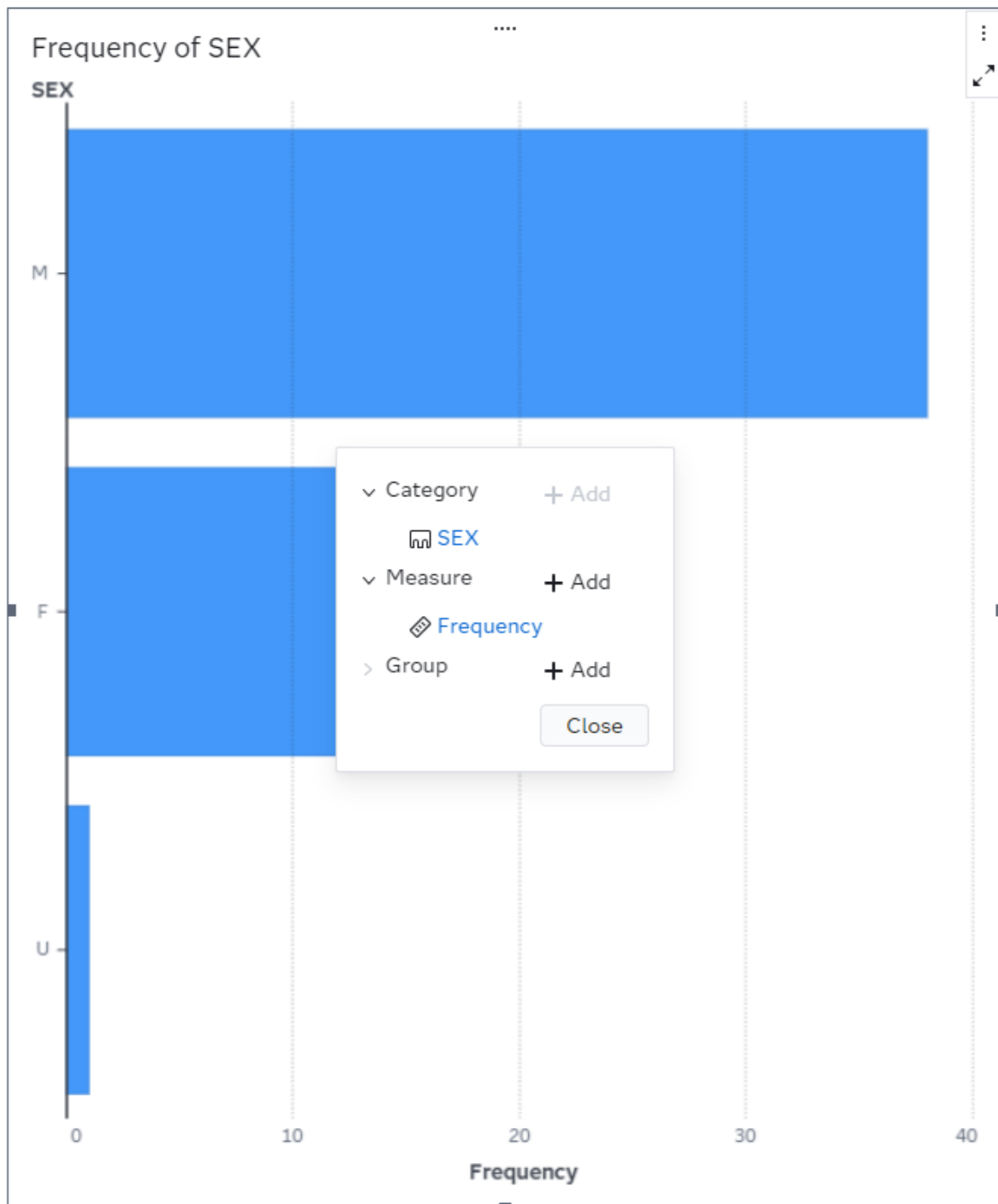
Step 3: Select **Objects** and drag & drop the **Bar chart** to right of the Pie chart



Step 4: Select Assign data



Step 5: Select the variable **SEX** as **Category**



Step 6: Select Options  on the right toolbar and change the **direction** of the Bar chart from

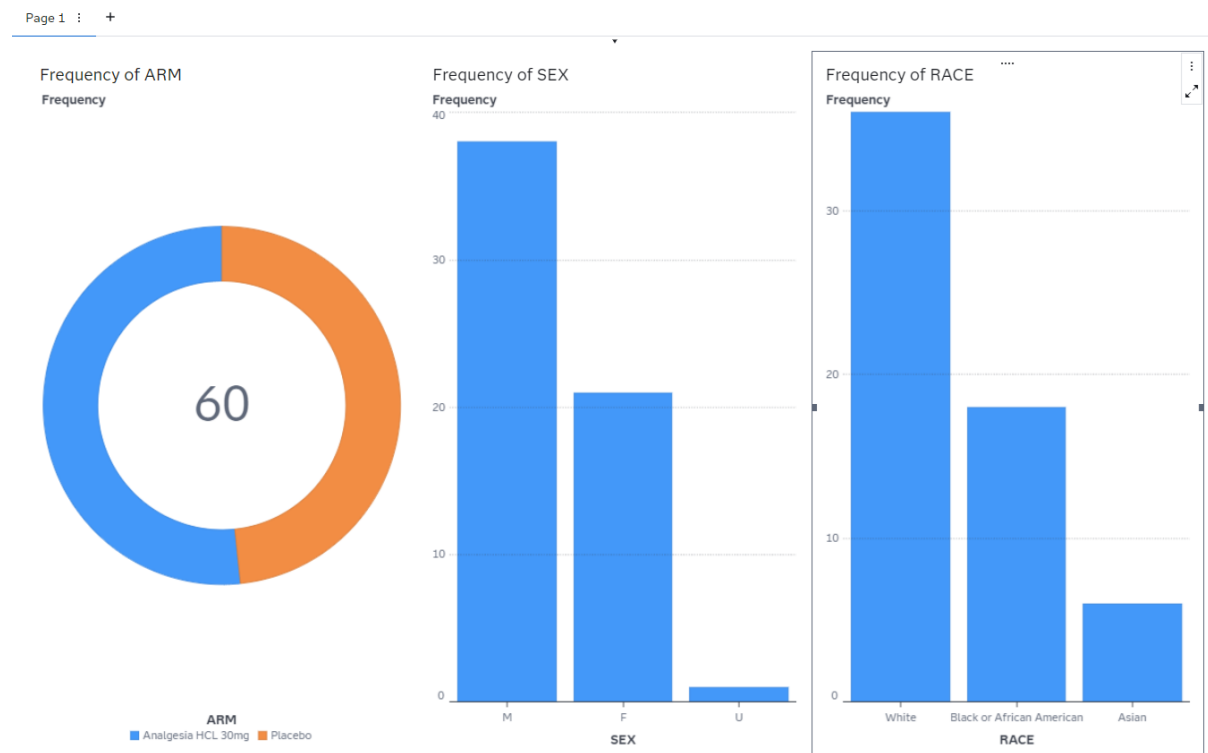
Direction:



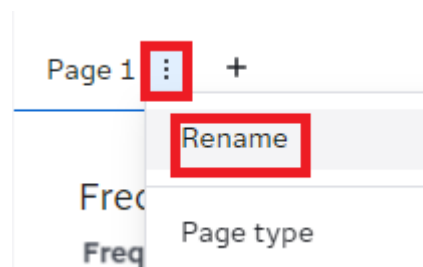
horizontal to **vertical**

Step 7: Select the **Options** menu and **Duplicate** the **Bar** chart

Step 8: Right mouse click inside the new Bar chart and select **Replace data SEX to RACE**



Step 9: Select the Page 1 menu Rename Page 1 to Demographic Dashboard



Step 10: Select **Data** and change the **Classification** from **SITEID** from Measure to **Category**

Measure

AGE

Frequency

lat

long

☐ SITEID

Name:

SITEID

Classification:

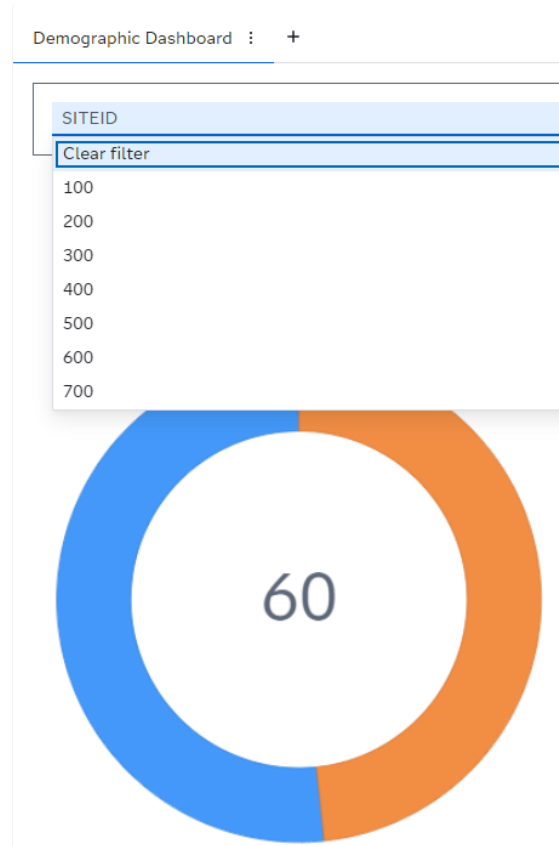
Measure

Category

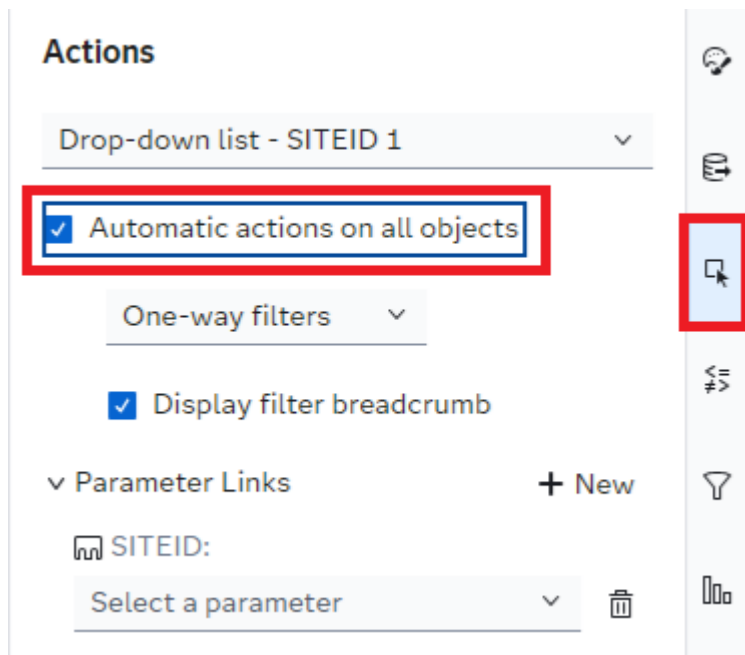
Geography

Measure

Step 11: Drag and drop a **Drop-down-List** to the **top of the page** and assign **SITEID** as **Category**



Step 12: Select **Actions**  and **Automatic actions on all objects**



Step 13: Select different SITIDs

Step 14: Select **Clear filter** for the Drop-down-list

Step 15: Select **Data** and select **"New custom category"** with a right mouse click on **AGE**

Step 16: Generate 3 groups

Young	0..39
Mid Age	40..49
Older People	50..120

New Custom Category ×

Name: Based on: Group by:

Value Groups + Add group ⋮

Group 1 🗑️

Name:

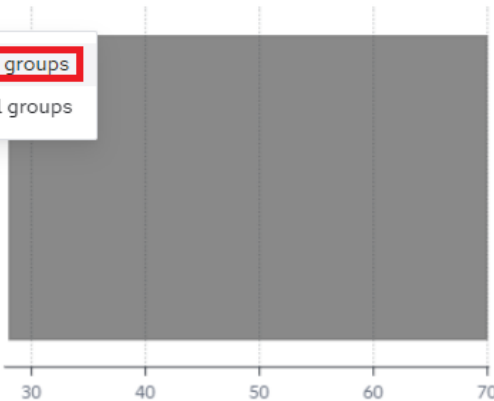
Intervals: + Add

Remaining Values ⌵

Generate groups

Delete all groups

Other



OK

Cancel

Step 17: Select OK

Generate Groups ×

Name:

Number of groups:

OK Cancel

Step 18: Select OK

Value Groups + Add group ⋮

Young 🗑️ ⬆️

Name: Young

Intervals: + Add

0 $\leq x \leq$ 39 🗑️

Mid Age 🗑️ ⬆️

Name: Mid Age

Intervals: + Add

40 $\leq x \leq$ 59 🗑️

Older People 🗑️ ⬆️

Name: Older People

Intervals: + Add

60 $\leq x \leq$ 120 🗑️

Remaining Values ⬇️

Step 19: Select **OK**

Step 20: Replace **SEX** to **Age Category** for the second Bar chart

Step 21: Select **Data** and select **Custom sort** with right mouse click on **Age Category** and sort the groups in the following sequence

Young, Mid Age, Older People

Step 22: Right mouse click on the **second Bar chart** and select **Sort – Age Category: Ascending**

Step 23: Add a **Button Bar** right to the **Drop-down-list** and assign **RACE** as **Category**

Step 24: Add a **List table** at the bottom of the page and assign

USUBJID, SEX, AGE, RACE and ARM

to the List table

Step 25: Define a Display Rule  on SEX for the first Bar chart

×

New Display Rule

SEX

↓

↑



+

 M

 F

☒ Other 

OK

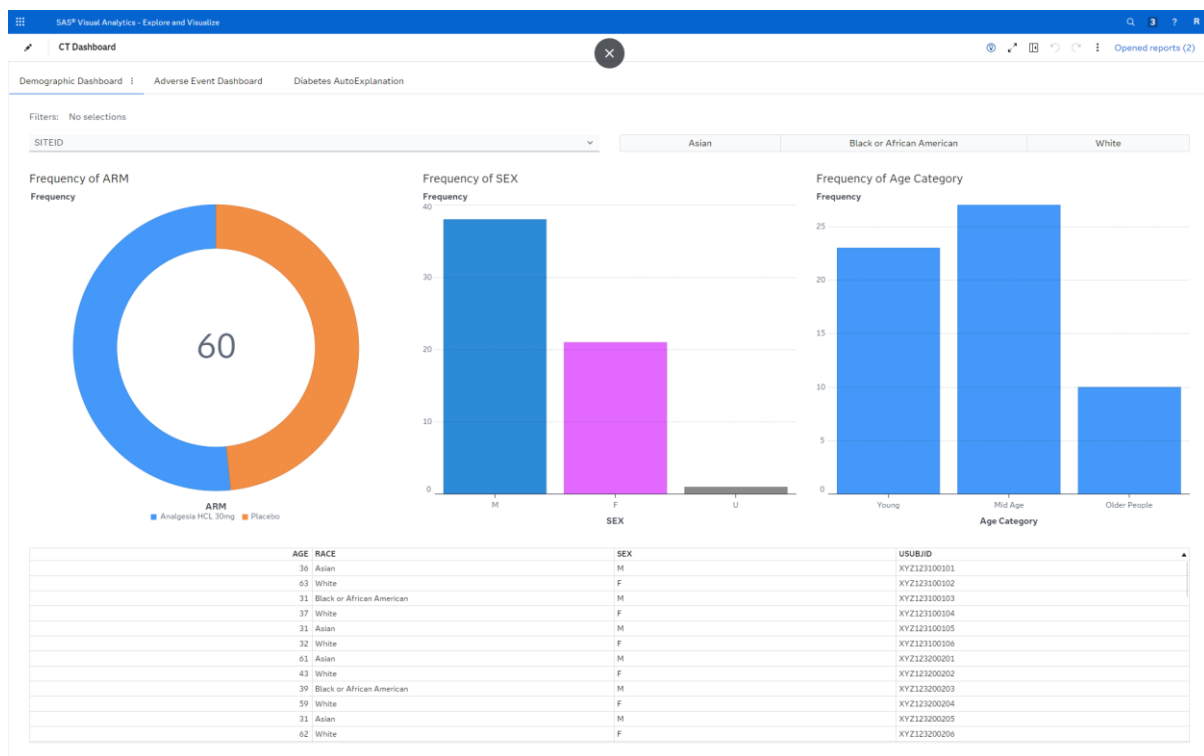
Cancel



Step 26: Change titles for all objects

- Select a SITE
- Select a RACE
- Frequency by ARM
- Split by Gender
- Age Category Coverage
-

Step 27: Disable Pie label and y Axis label



Step 28: Save the report  under My Folder “Clinical Trial Dashboard”

Adverse Event Dashboard

Step 1: Add **AE_DM** data to the report

Step 2: Add a new page “**Adverse Event Dashboard**”

Step 3: Create a **List table** with the following variables

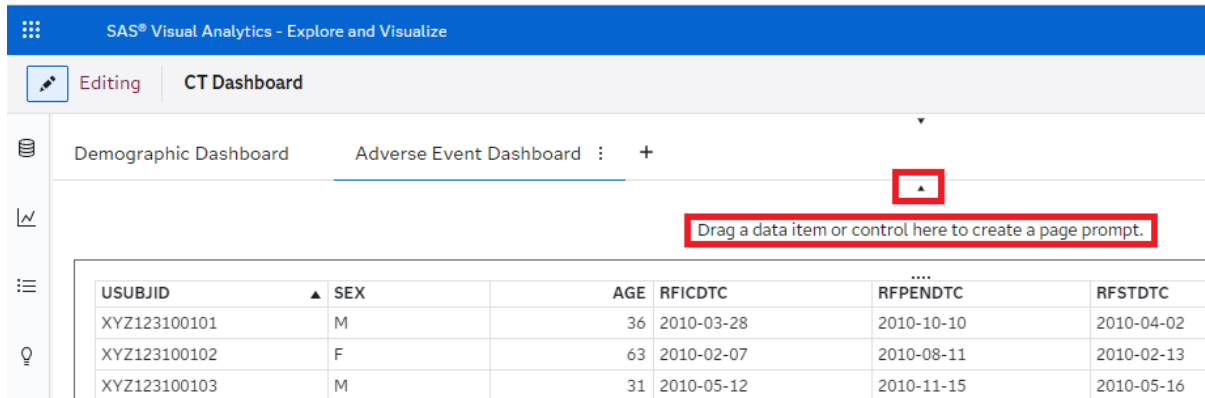
USUBJID, SEX, RFENDTC, RFICDTC, RFPENDTC, RFSTDTC, RFXENDTC, RFXSTDTC, AEBODSYS

Step 4: Change the **Classification** from **AGE** to **Category**

Step 5: Add Age to the List table and shift it after SEX

Step 6: Change the **Classification** of **SITEID** to **Category**

Step 7: Drag & drop **SITEID** into the **Page control area**. Click on the **black triangle** and drag & drop **SITEID** onto the text “**Drag a data item or control here to create a page prompt**”



USUBJID	SEX	AGE	RFICDTC	RFPENDTC	RFSTDTC
XYZ123100101	M	36	2010-03-28	2010-10-10	2010-04-02
XYZ123100102	F	63	2010-02-07	2010-08-11	2010-02-13
XYZ123100103	M	31	2010-05-12	2010-11-15	2010-05-16

Step 8: Change the created **Drop-down-list** to a **Button bar**



Set **title** to “**Site Number**”

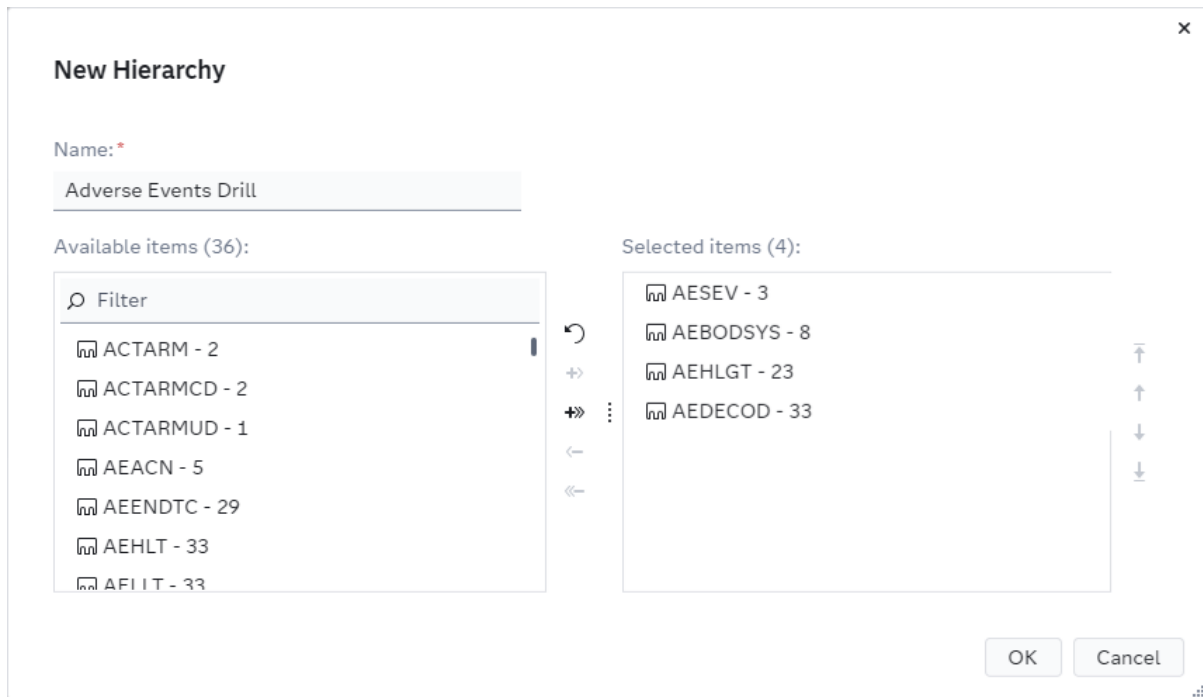
Step 9: Drag & drop **AESER** to the right side of the Button Bar. Set **title** to “**Is AE Serious?**”

Step 10: Create a new hierarchy “**Adverse Events Drill**” with the levels

AESEV, AEBODSYS, AEHLGT and AEDECOD

- Select **Data** and then + **New data item**
- Select **Hierarchy**

c) Set the levels as follows



New Hierarchy

Name: *
Adverse Events Drill

Available items (36):

- Filter
- ACTARM - 2
- ACTARMCD - 2
- ACTARMUD - 1
- AEACN - 5
- AEENDTC - 29
- AEHLT - 33
- AEHLT - 33

Selected items (4):

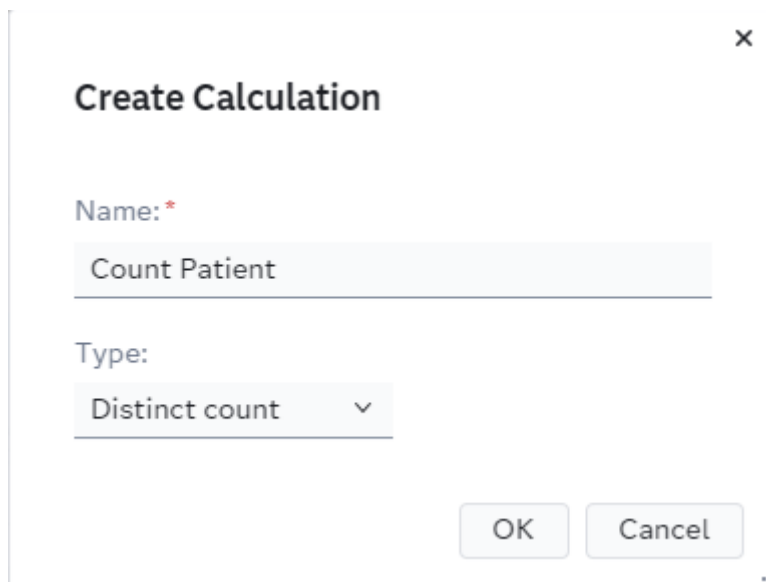
- AESEV - 3
- AEBODSYS - 8
- AEHLGT - 23
- AEDECOD - 33

OK Cancel

Step 11: Drag & drop the **Hierarchy** at the **top of the List table**

Step 12: Change the **Bar chart** to a **Pie chart**

Step 13: Create a “**New calculation**” **Distinct count** with the named “**Count Patient**” for variable **USUBJID**



Create Calculation

Name: *
Count Patient

Type:
Distinct count

OK Cancel

Step 14: Replace the Measure variable **Frequency** to **Count Patient** for the **Pie chart**

Step 15: Change **title** to **Count Patient**

Step 16: Under Options **deselect Pie label**, **select Actual values**, **select Category labels** and set the **Legend** to **Visibility Off**, set **Data label location** to **Outside**

Step 17: Select **Y** for **AE serious**

Step 18: Drill down into different segments of the Pie chart

Step 19: Drag & drop **AEREL** to the right side of the report

Step 20: Define a **Filter** on **ACTARMCD** for the Bar chart and **deselect ALG123**, **deselect Including missing values**

Step 21: **Deselect Axis label** and **Tick values** for the **X-Axis**

Step 22: **Deselect Axis label** and **Axis line** for the **Y-Axis**, set **Legend Visibility** to **Off**

Step 23: Select **Data labels** and set **Text style** to **Bold, 20, blue**

Step 24: Set **title** to **Impact for Placebo**

Step 20: **Duplicate** the **Bar chart** and move it under the first Bar chart

Step 21: Change **title** to **Impact for Product**

Step 22: Change **Filter**, select only **ALG 123**

Step 23: Change the size of the Bar charts and the Pie chart

Step 24: Explore the data

Step 25: Define a **Display Rule** on **Frequency** for the first Bar chart with

×

New Display Rule

Frequency

Label:

Operator:

>

Value:

3

Style area:

Background

Style:



Transparency:

0.8

☒ Allow alerts for this rule

Recipients:

Ralf.Storch@sas.com ×  

Notification frequency: Interval (days):

Daily 1

OK Cancel



Step 26: Define a Display Rule on AEREL for the second Bar chart

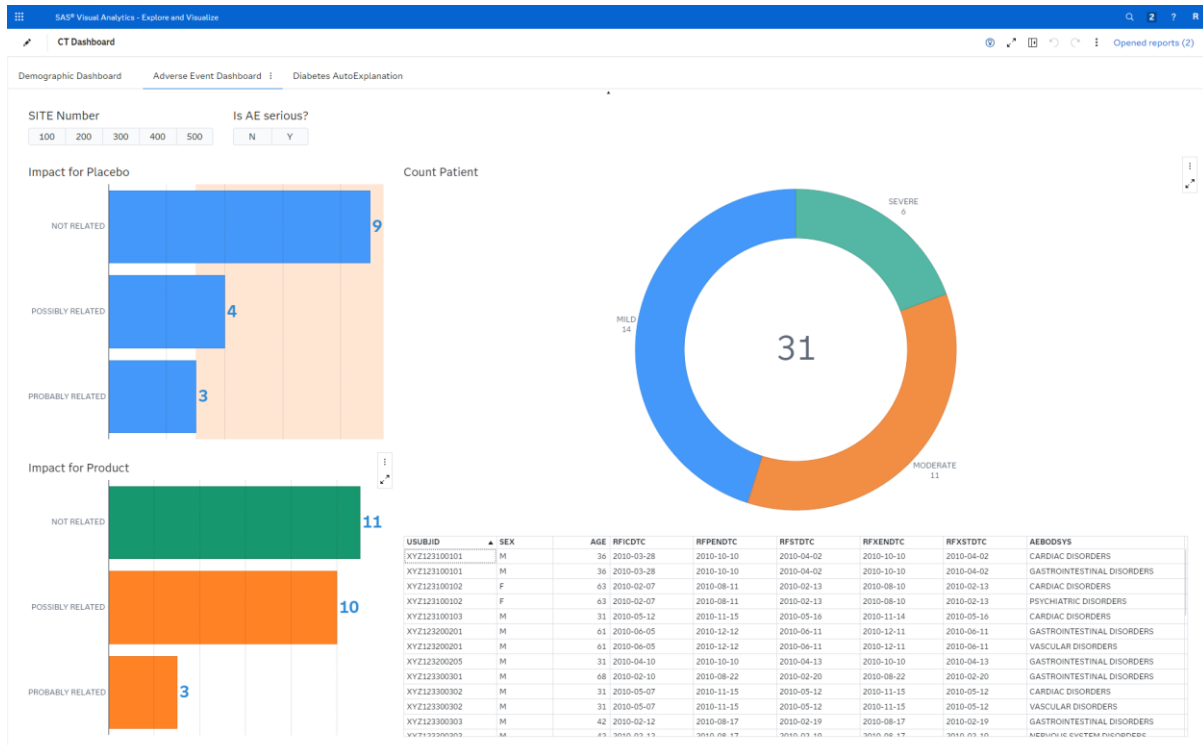
New Display Rule

AEREL

☒ NOT RELATED

☒ Other

OK Cancel



Step 27: Save the report

Real World Data Exploration - Example

Auto Explanation for SASLIN on Diabetes

Step 1: Create a **new page** in the same report and **rename** the page to **Diabetes AutoExplanation**

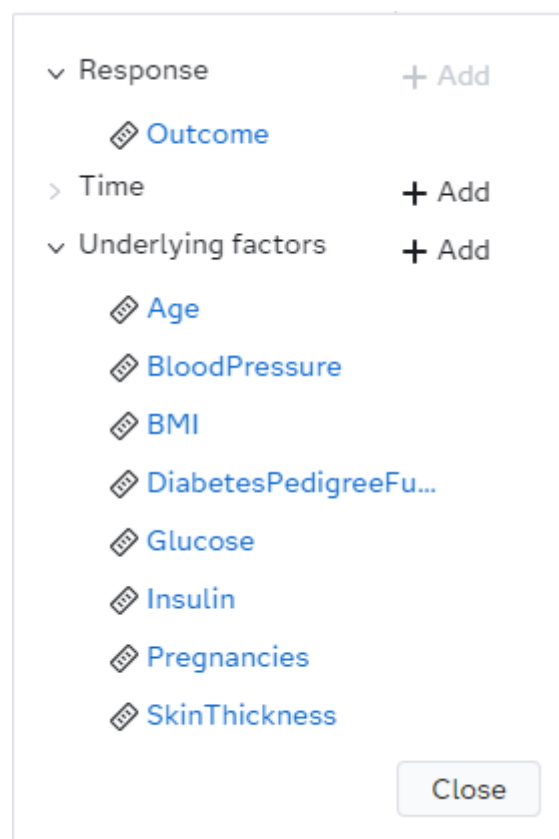
Step 2: Add the **Diabetes** data to the report

Step 3: Change the **Classification** of **Outcome** to **Category**

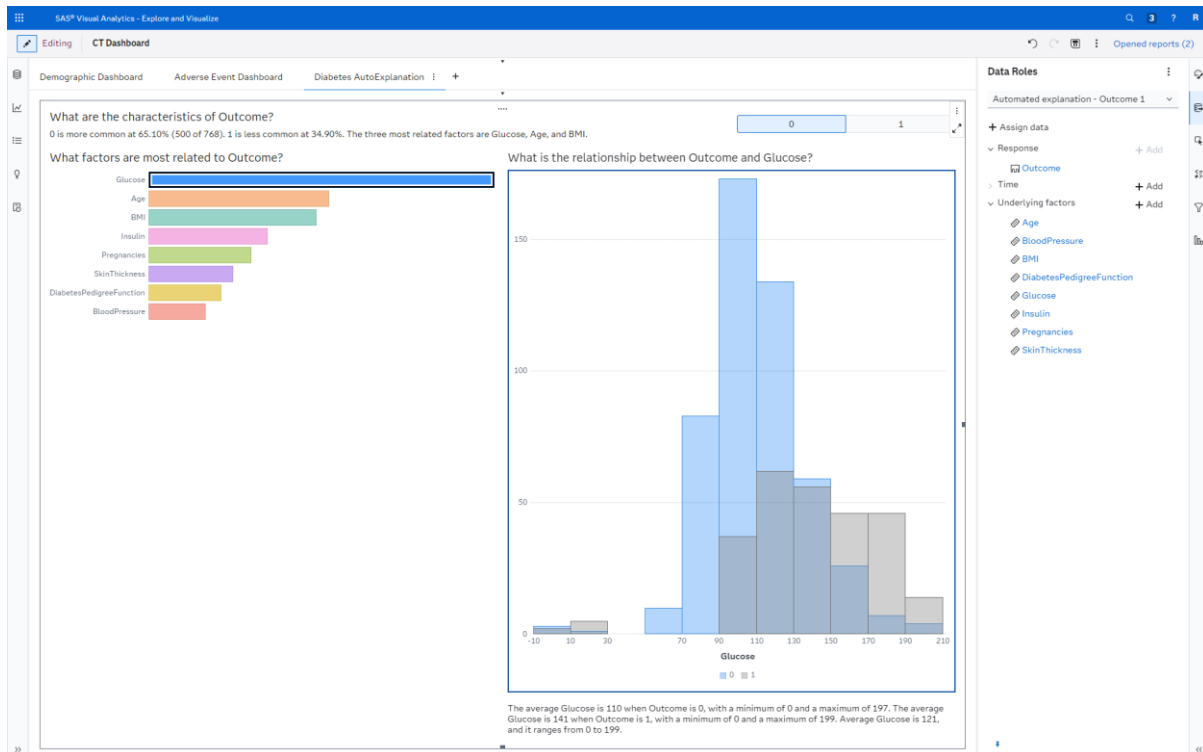
Step 4: Drag & drop “**Automated Explanation**” from **Objects** to the canvas

Step 5: For the **Data Roles** for this “**Automated Explanation**” choose

- **Outcome** for **Response**
- **All other variables** are automatically selected for **Underlying factors**



Once you click on “**Close**”, your Automated Explanation object comes up after a while as follows



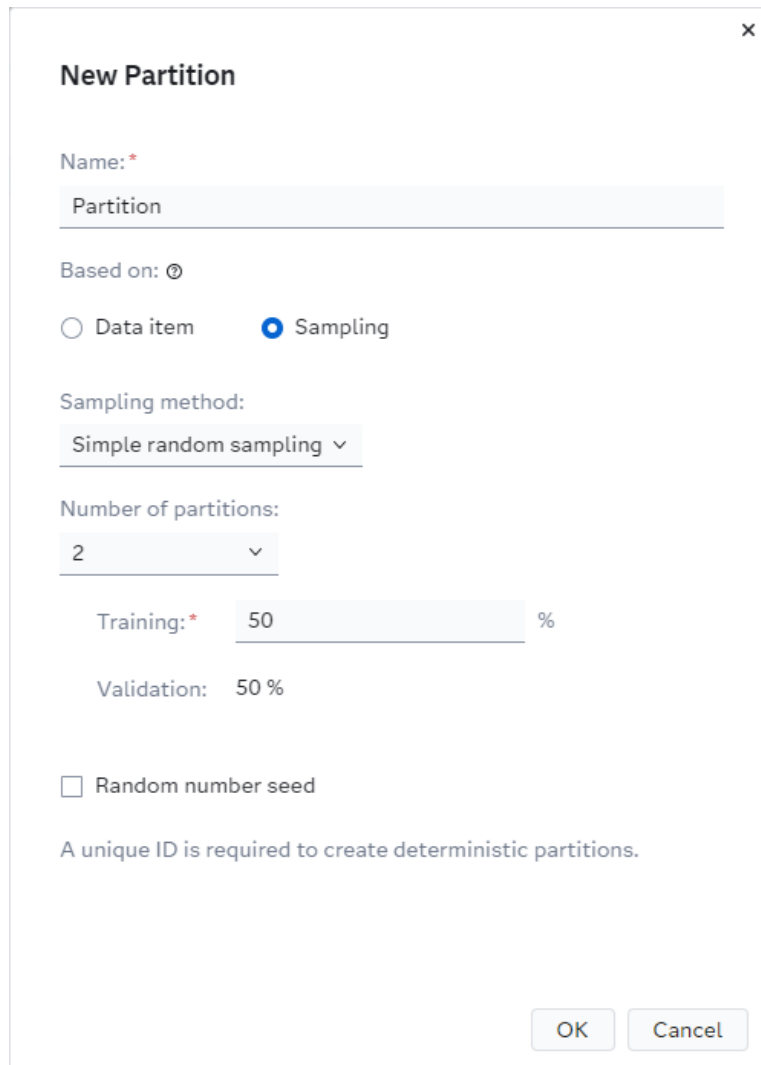
You will notice the most important factor (related to response variable Outcome).

You can select the other factors to see how the graph and data / explanation is changing dynamically.

Prediction on Diabetes - Modeling

Step 1: Partition the data with Sampling method

- **Training 50%**
- **Validation 50%**



New Partition

Name: *

Partition

Based on: Ⓢ

☐ Data item ☒ Sampling

Sampling method:

Simple random sampling ▾

Number of partitions:

2 ▾

Training: * 50 %

Validation: 50 %

☐ Random number seed

A unique ID is required to create deterministic partitions.

OK Cancel

Step 2: Create a new page **Modeling**

Step 3: Drag & drop a **Logistic regression** on the canvas

Step 4: For the **Data Roles** for this “**Logistic regression**” choose

- **Outcome** as **Response** variable
- **Partition** as **Partition ID**
- and **all other variables** as **Continuous effects**

Data Roles ⋮

Logistic regression - Outcome 1 ▾

+ Assign data

✓ Response + Add

Outcome

✓ Continuous effects + Add

Age

BloodPressure

BMI

DiabetesPedigreeFunction

Glucose

Insulin

Pregnancies

SkinThickness

> Classification effects + Add

> Interaction effects + Add

✓ Partition ID + Add

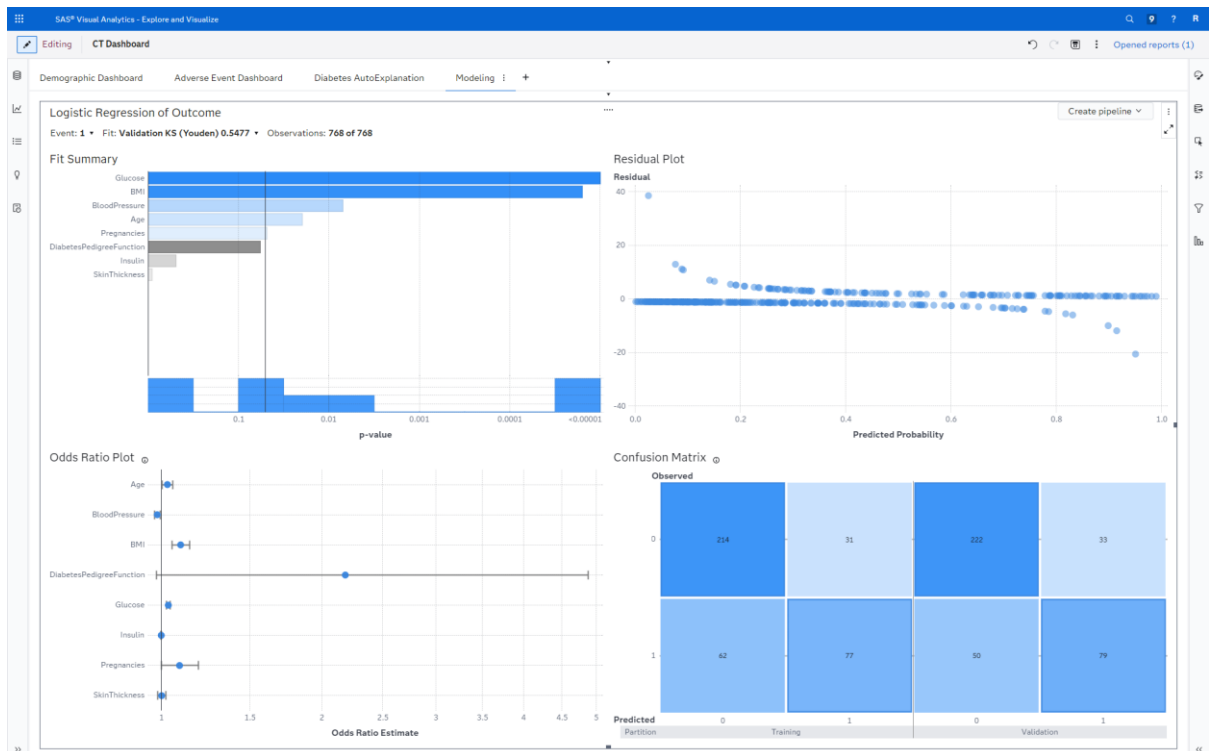
Partition

> Group by + Add

> Frequency + Add

> Weight + Add

> Offset + Add



Step 5: Select Fit Statistic for Training and Validation

Step 6: Duplicate the **Logistic Regression** as **Forest** and move to the left side of the report page

Step 7: Duplicate the **Forest** as **Decision Tree** and move to the left side of the report page

Step 8: Change the **Plot layout** to **Stack**



Step 9: Select **Model comparison** and drag & drop it on the + right to page Modeling to create a **new page** and rename the page to **Model Comparison** and select **all models**

Add Model Comparison

Data:

DIABETES

Partition:

Partition

Response:

Outcome

Event level:

1

Available models:

[Clear selection](#)

- ☒ Forest - Outcome 1
- ☒ Decision tree - Outcome 1
- ☒ Logistic regression - Outcome 1

OK

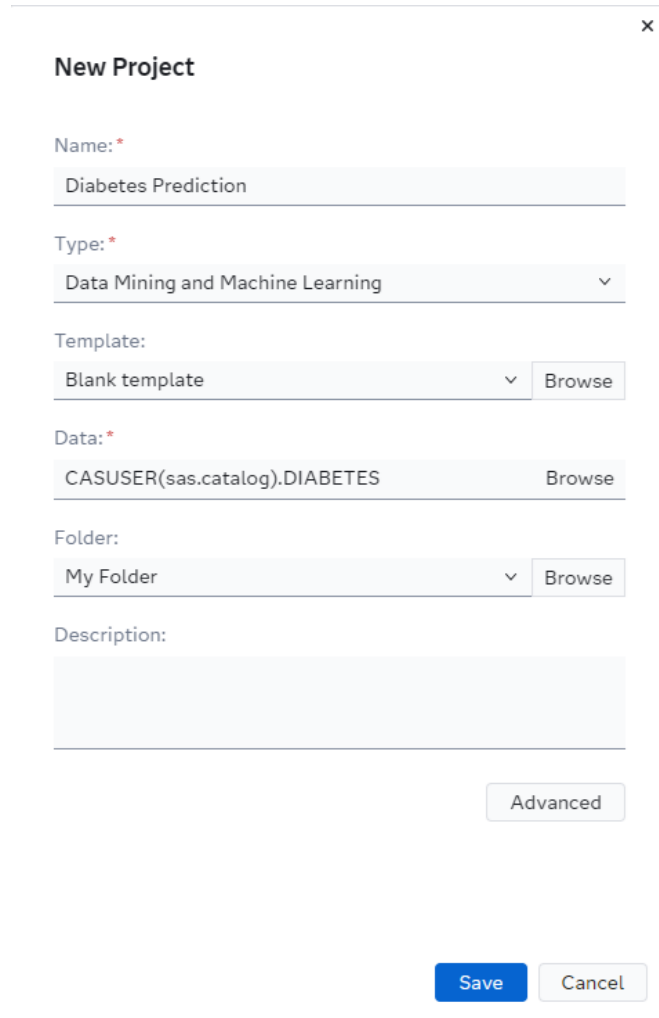
Cancel



Prediction on Diabetes – Advanced Modeling, Pipeline Building

Step 1: Open the **Model Studio** via the **Applications** menu and select **Build Models**

Step 2: Select **New project** and set the **Name** to “**Diabetes Prediction**” and select the **DIABETES** data



New Project

Name: *
Diabetes Prediction

Type: *
Data Mining and Machine Learning

Template:
Blank template Browse

Data: *
CASUSER(sas.catalog).DIABETES Browse

Folder:
My Folder Browse

Description:

Advanced

Save Cancel

Step 3: Select **Save**

Model Studio - Build Models

Projects > Diabetes Prediction

Diabetes Prediction

Data Pipelines Pipeline Comparison Insights

You must assign a variable with the role of Target to run a pipeline.

Variable Name	Label	Type	Role	Assess for Bias	Level	Order	Comment	Number of Levels
Age		Numeric	Input		Interval	Default		52
BloodPressure		Numeric	Input		Interval	Default		47
BMI		Numeric	Input		Interval	Default		248
DiabetesPedigreeFunction		Numeric	Input		Interval	Default		>254
Glucose		Numeric	Input		Interval	Default		136
Insulin		Numeric	Input		Interval	Default		186
Outcome		Numeric	Input		Binary	Default		2
Pregnancies		Numeric	Input		Nominal	Default		17
SkinThickness		Numeric	Input		Interval	Default		51

DIABETES

Columns:
9

Rows:
768

Label:
(not available)

Location:
cas-shared-default/CASUSER(raif.storch@sas.com)

Step 4: Select variable **Outcome** and define it **Role** as **Target**

Outcome

Role:

Target

Level:

Binary

Specify the Target Event Level

Order:

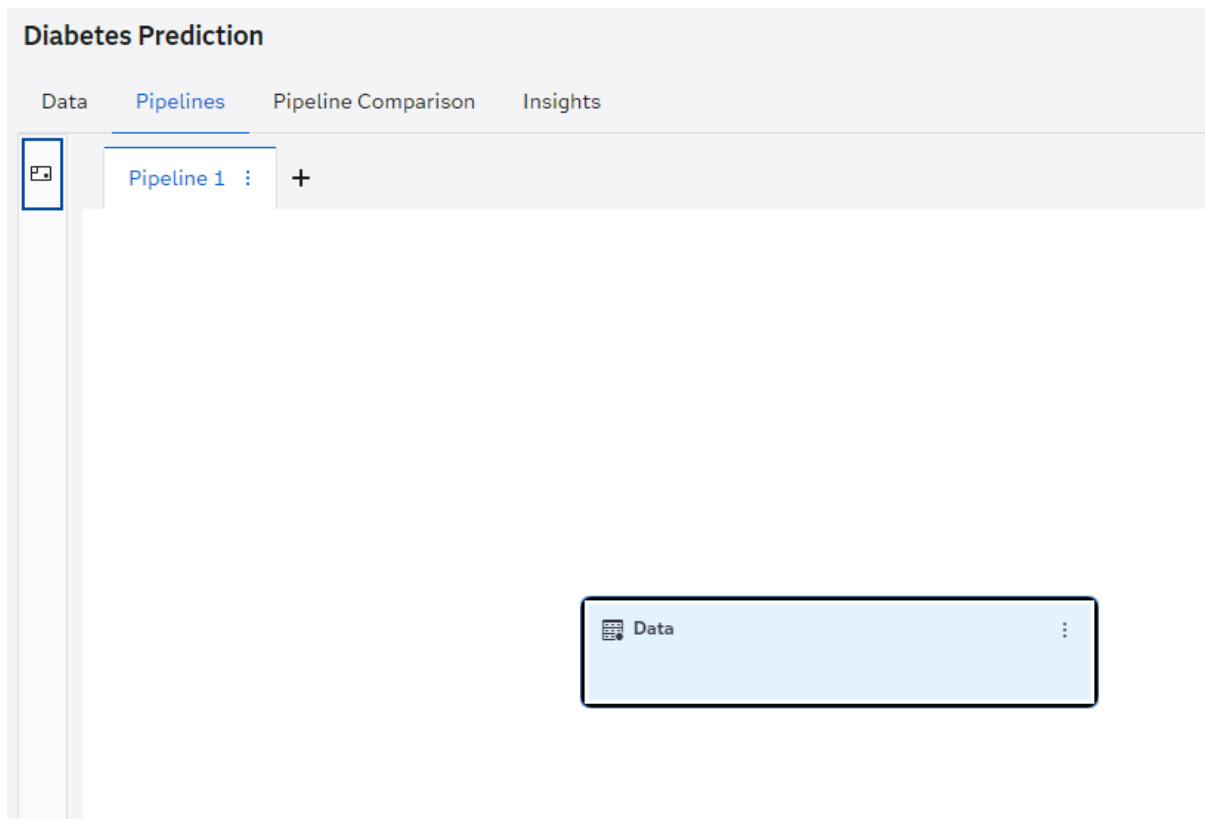
Default

Transform:

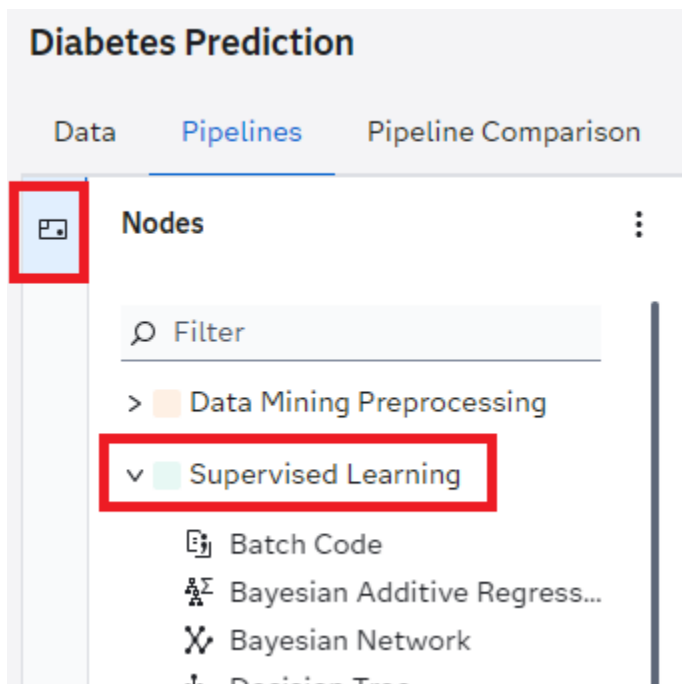
Impute:

Step 5: Select variable **Pregnancies** and define it **Level** to **Interval**

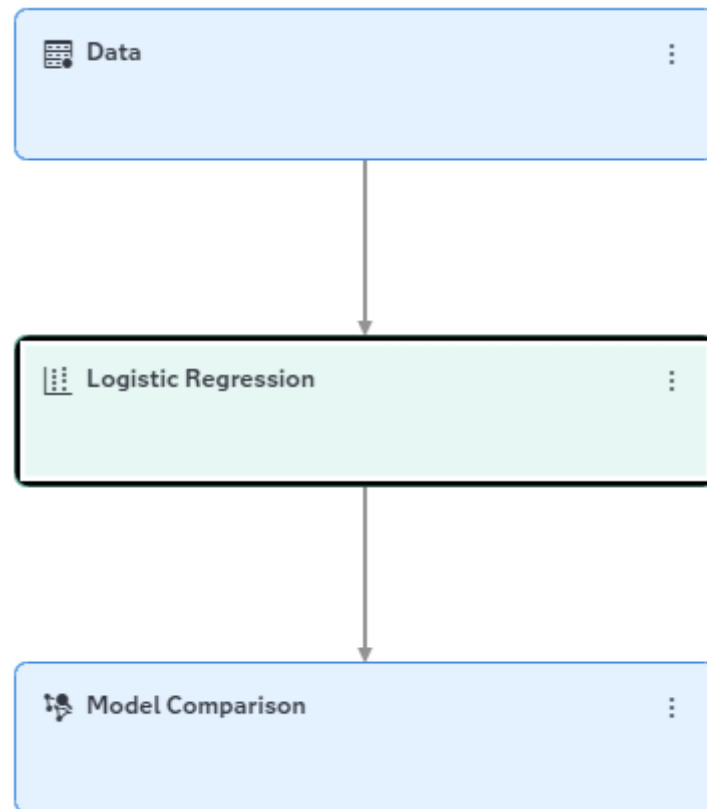
Step 6: Select the tab **Pipelines**



Step 7: Open **Nodes** and open **Supervised Learning**



Step 8: Drag & drop the **Logistic Regression** on the **Data** node



Step 9: Open **Nodes** and open **Misellaneous**

Step 10: Drag & Drap **Open Source Code** node on the **Data** node

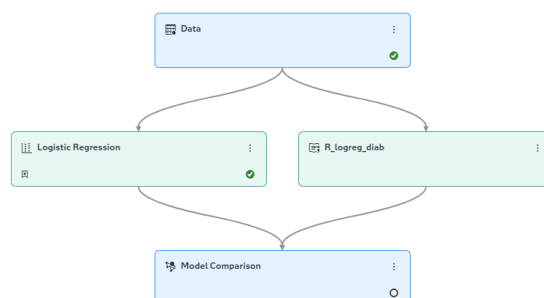
Step 11: Select **R** as Language and select the button “**Open code editor**”

Step 12: Past the **R code (R Code.txt)** from Github into the code editor

Step 13: **Save** and **Close**

Step 14: Select the node menu and **rename** the **node** to **R_logreg_diab**

Step 15: Select the node menu and **Move** the node to **Supervised Learning**



Step 16: Run pipeline

Step 17: Right mouse click on node R_logreg_diab and select Results

Step 18: Select Assessment

Step 19: Close and Close

Step 20: Rename Pipeline to Basic

Step 21: Duplicate Pipeline and rename it to Intermediate

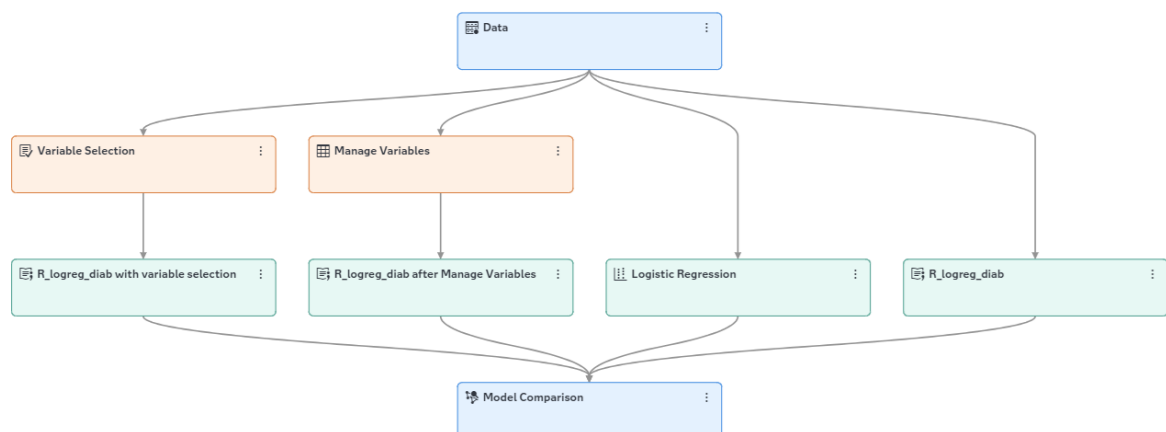
Step 22: Open node and Data Mining Preprocessing and drag & drop Variable Selection and Manage Variables on the Data node. Close the dialog box.

Step 23: Copy and paste the R_Logreg_diab node to the node Variable Selection

Step 24: Copy and paste the R_Logreg_diab node to the node Manage Variables

Step 25: Rename R_Logreg_diab (1) to R_Logreg_diab with variable selection

Step 26: Rename R_Logreg_diab (2) to R_Logreg_diab after manage variables



Step 27: Run pipeline

Step 28: Open Results

Step 29: Duplicate Pipeline and rename it to Advanced

Step 30: Open node and Miscellaneous and drag & drop SAS Code on the Data node.

Step 40: Past the SAS Code (SAS Code.txt) from Github into the code editor

Step 41: Save and Close

Step 42: Drag & Drap Variable Selection node on the SAS Code node

Step 43: Drag & drop

Decision Tree, Forest, Logistic Regression
on the **Variable Selection** node

Step 44: Copy and paste the **R_Logreg_diab** node to the node **Variable Selection**

Step 45: Run pipeline

Step 46: Open Results

Step 47: Select **Assessment**

Step 48: Open **Results** for **Model Comparison**

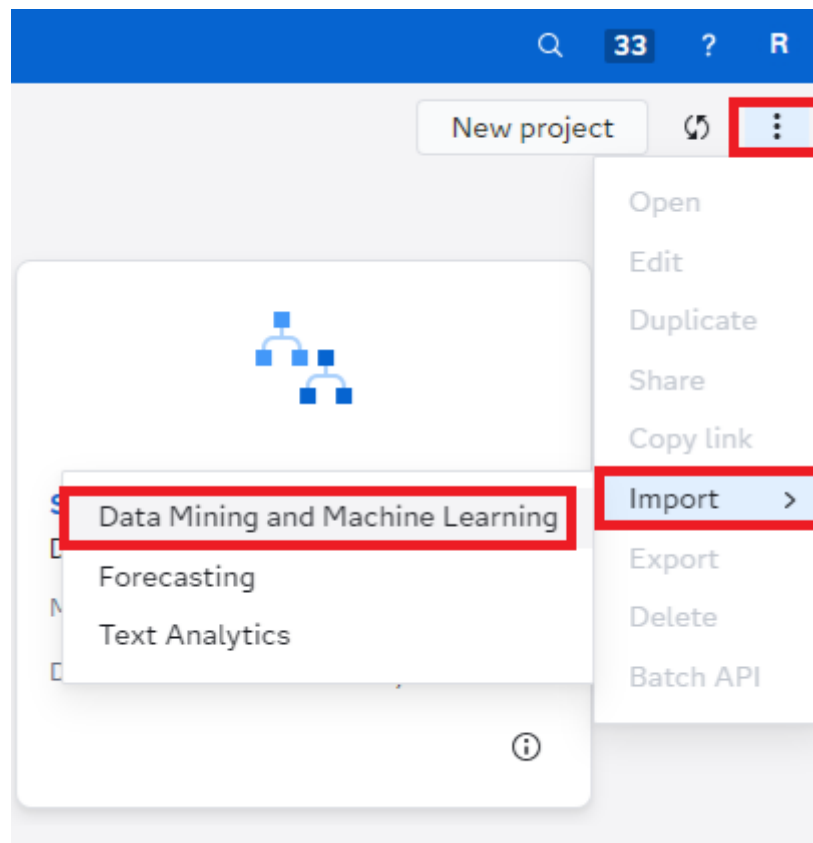
Step 49: Select tab **Pipeline Comparison**

Step 50: Create PDF report

Step 51: Open PDF report

Step 52: Go to the start page of the Model Studio and

Step 53: Import the original project “**Diabetes Prediction (ORG).zip**” from github to compare your results. Select **Data Mining and Machine Learning**



Select your **DIABETES** data in your CASUSER(<your Email>) library.

×

Import Data Mining Project

File (zip): *

Diabetes Prediction (ORG).zip
Browse

Data: *

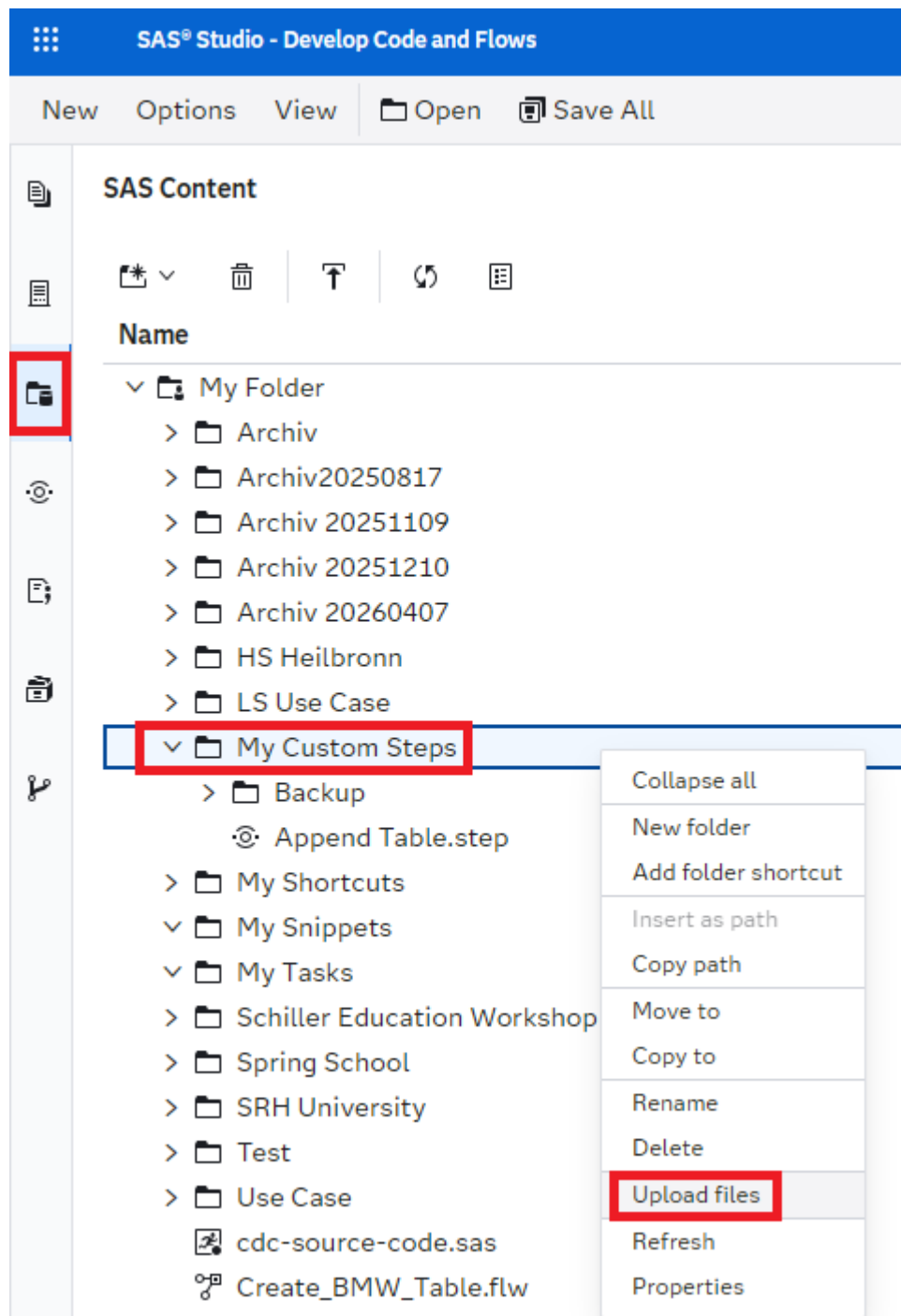
CASUSER(ralf.storch@sas.com).DIABETES
Browse

Import
Cancel

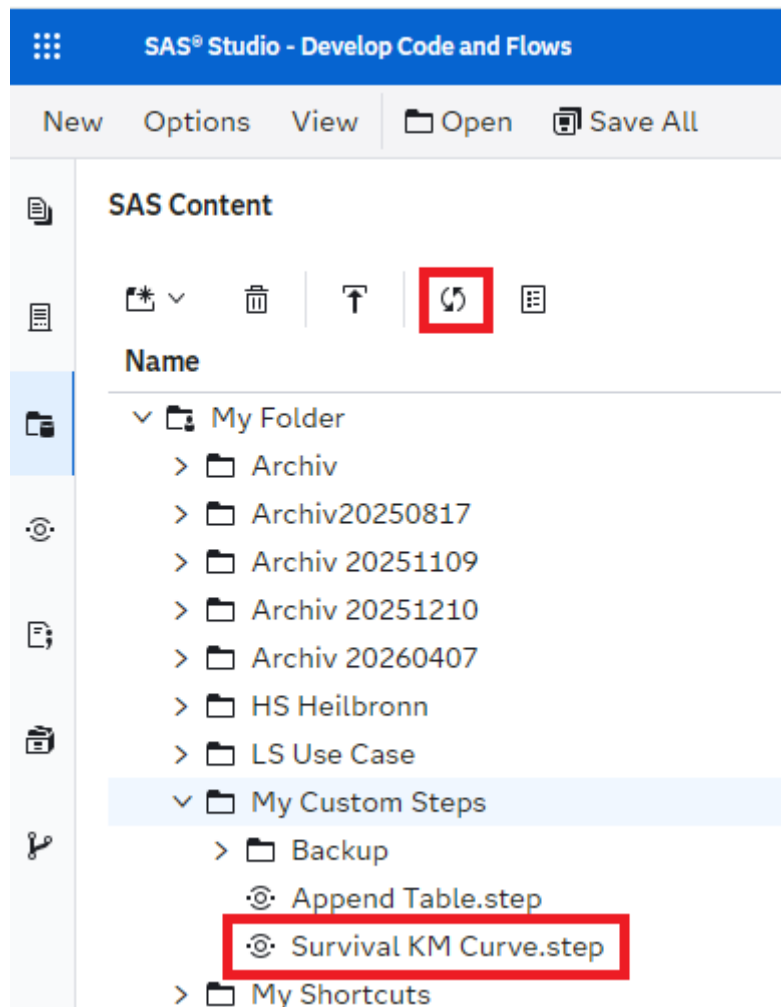
Survival Analysis using LowCode Approach

Step1: Select the **Applications** menu and select **Develop Code and Flows** to open **Open SAS Studio**

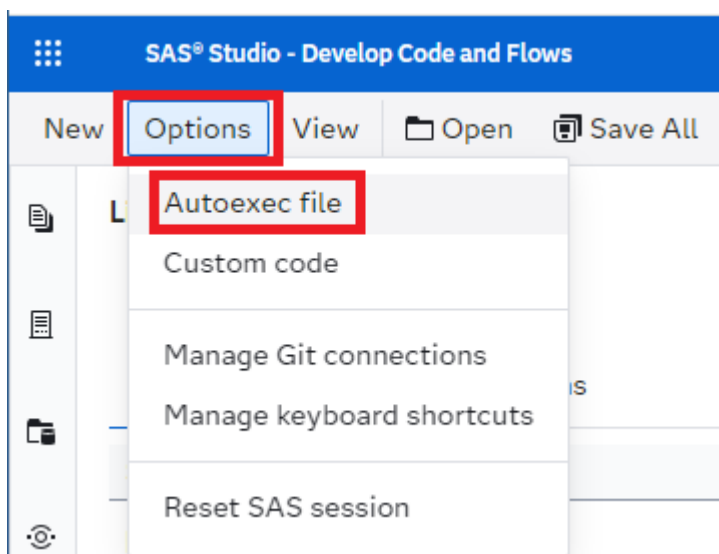
Step 2: Select **SAS Content**  in the left toolbar and open the subfolder **"My Custom Steps"** and with a right mouse click open the **context menu** and select **Upload files** and upload the file **Survival KM Curve.step** from Github



Step 3: Refresh  the list



Step 4: Select **Options** and then **Autoexec file**



Step 5: Copy and paste the **3 commands** from the file **Autoexec.txt** on Github into the editor

Step 6: Select **Run** and **Save**

Autoexec File

Specify the SAS statements to run when application starts. To validate the syntax, click Run.

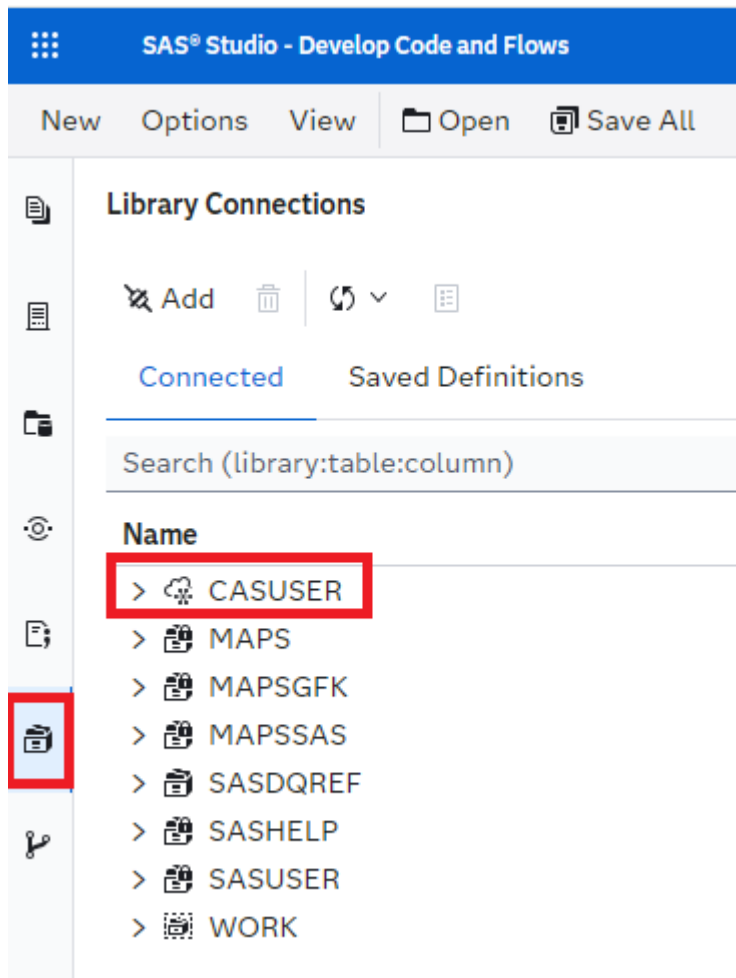
Include in background submits ☒

Code	Log
1	
2	<code>cas mySession terminate;</code>
3	
4	<code>cas mySession sessopts=(timeout=3600);</code>
5	
6	<code>libname casuser cas caslib=casuser;</code>
7	
8	

Run

Save Cancel

Step 7: Select **Library Connections** in the left toolbar and open **CASUSER**



Step 8: Select **+** beside the tab **Start Page** and select **Flow**

Step 9: Drag & Drop the data ADTTE under CASUSER to the canvas of the flow

Step 10: Select **Steps**  in the left toolbar

Step 11: Select the tab **Custom Steps**

Step 12: Drag & drop the **Survival KM Curve** step to the flow and link the **table node** to **Survival KM Curve node**



Step 13: Select the Survival KM Curve node to see the properties at the bottom of this node

Step 14: Set the following variables

 Survival KM Curve

Page 1 Node Notes

Select Time Variable *		
⊕ AVAL		
Select Status Variable *		
⊕ CNSR		
Select Censored Value		
0		
Select Group Variable *		
⚠ TRTP		
Enter the units of Time		

Step 15: Run the flow or the Survival KM Curve node

Step 16: Select **Submitted Code and Results**

Step 17: Open in a browser tab, Summary